

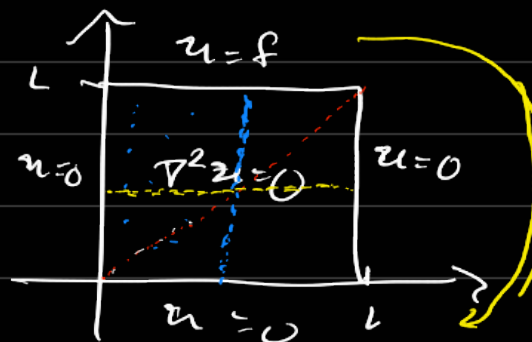
$$\frac{a_0}{2} L \delta_{m,0} + \sum_{n=1}^{\infty} a_n \frac{L}{2} \delta_{nm} = \dots$$

$$= \sum_{n=0}^{\infty} \frac{a_n}{2} L \delta_{nm}$$

From yesterday: General Soln

$$u(x,y) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) \sinh\left(\frac{n\pi y}{L}\right)$$

$$b_n = \frac{2 \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx}{L \sinh(n\pi)}$$



Symmetries of Laplace Eq.:

$$* \nabla^2 u = u_{xx} + u_{yy} = 0$$

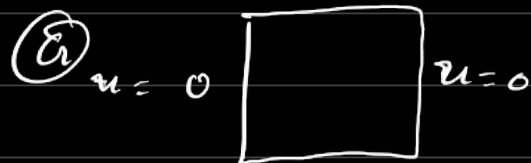
$$* (x \leftrightarrow y) *$$

$$x=y$$

$$u_{yy} + u_{xx} = u_{xx} + u_{yy}$$

$$\begin{cases} \hat{x} = y \\ \hat{y} = x \end{cases}$$

$$u_{\hat{y}\hat{y}} + u_{\hat{x}\hat{x}} = 0$$



$$u=f$$

$$\hat{x} = x$$

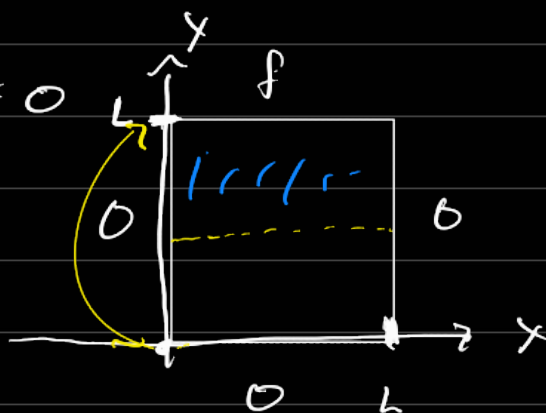
$$\hat{y}: y=L: \text{ want } \hat{y}: 0$$

$$\hat{y} = L-y$$

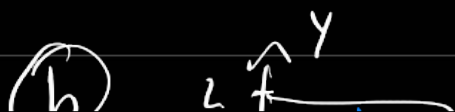
$$\hat{y}(y) = L-y$$

$$\hat{y}(L) = 0$$

$$\hat{y}(0) = L$$



$$\begin{cases} \hat{x} = x \\ \hat{y} = L-y \end{cases}$$



$$\hat{x} = 1-x$$

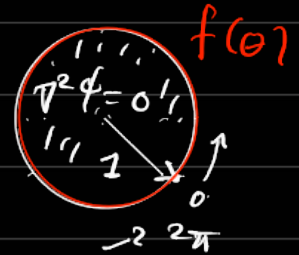


Laplace in Polar:  $u = \phi$

$$\nabla^2 \phi = \phi_{xx} + \phi_{yy} = \phi_{rr} + \frac{1}{r} \phi_r + \frac{1}{r^2} \phi_{\theta\theta} = 0$$

change of coordinates

$$(x, y) \mapsto (r, \theta)$$



$$\phi_{rr} + \frac{1}{r} \phi_r + \frac{1}{r^2} \phi_{\theta\theta} = 0$$

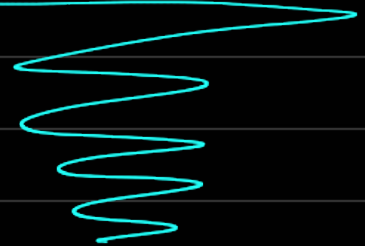
$$\bullet \quad 0 \leq r < 1 \quad \leftarrow$$

$$0 \leq \theta < 2\pi$$

$$\ast \quad \phi(1, \theta) = f(\theta) \quad \leftarrow$$

$$\ast \quad \phi \text{ } 2\pi\text{-periodic in } \theta$$

$\phi(0, \theta)$  is Bounded



Solve: Sep. of variables:

$$\phi(r, \theta) = F(r)G(\theta)$$

plug this into  $\phi_{rr} + \frac{1}{r} \phi_r + \frac{1}{r^2} \phi_{\theta\theta} = 0$   
will get

$$\frac{r^2 F'' + r F'}{F} = - \frac{G''}{G} = + \lambda^2 \quad (2)$$

• This case is different to the others: Not linear

• Gives two ODEs:

$$G''(\theta) = -\lambda^2 G(\theta)$$

$$G(\theta) = A \cos(\lambda \theta) + B \sin(\lambda \theta), \quad \lambda \neq 0$$

$$\lambda = 0: \quad G''(\theta) = 0$$

$$G(\theta) = \tilde{A} \theta + \tilde{B}$$

$$G(0) = G(2\pi)$$

$$\lambda = 0: \quad G(0) = \tilde{B}$$

$$G(2\pi) = \tilde{A} 2\pi + \tilde{B}$$

$$A 2\pi + \tilde{B} = \tilde{B}$$

$$\tilde{A} 2\pi = 0 \Rightarrow \tilde{A} = 0$$

$$G_0(\theta) = \tilde{B} \quad (\text{constant})$$

$$\lambda \neq 0: G(\theta) = A \cos(\lambda \theta) + B \sin(\lambda \theta)$$

$$G(0) = A$$

$$G(2\pi) = A \cos(\lambda 2\pi) + B \sin(\lambda 2\pi)$$

$$A = A \cos(\lambda 2\pi) + B \sin(\lambda 2\pi)$$

$$\Rightarrow \sin(\lambda 2\pi) = 0$$

$$\lambda 2\pi = n\pi$$

$$\lambda 2\pi = \pi$$

$$\lambda = \frac{1}{2}$$

$$n = 1, 2, 3, \dots$$

$$\cos(\lambda 2\pi) = 1$$

$$\lambda 2\pi = 2\pi n \Rightarrow \lambda = n$$

$$\lambda = n, \quad n = 0, 1, 2, \dots$$

$$\begin{cases} G_0(\theta) = \text{const.} & \lambda = 0 \\ G_n(\theta) = A_n \cos(n\theta) + B_n \sin(n\theta), & \lambda \neq 0 \end{cases}$$

$$* \text{ "r" -variable: } \frac{r^2 F'' + r F'}{F} = \lambda^2 = n^2$$

$$\bullet \quad r^2 F'' + r F' - n^2 F = 0$$

$$F = r^a, \quad a = \pm n$$

$$(i) \quad n = 0$$

$$F_0(r) = C_0 + D_0 \log(r)$$

$$(ii) \quad n \neq 0$$

$$F_n(r) = C_n r^n + D_n r^{-n}$$

$$F_n(r) = \begin{cases} C_0 + D_0 \log(r), & n = 0 \\ C_n r^n + D_n r^{-n}, & n \neq 0 \end{cases}$$

look back  $0 \leq r < 1$ , and  $\phi(0, \theta)$  bounded

$$\lim_{r \rightarrow 0} F_0(r) = \lim_{r \rightarrow 0} (C_0 + D_0 \log(r))$$

$$= C_0 + D_0 \underbrace{\lim_{r \rightarrow 0} \log(r)}$$

$$= \pm \infty$$

• Can't have  $\infty$ ,  $F$  is Bounded, thus  $D_0 = 0$

$$F_n(r) : \quad r^{-n} \rightarrow \infty$$

Again  $D_n = 0$

$$F_n(r) = \begin{cases} C_0 & n=0 \\ C_n r^n & n \neq 0 \end{cases}$$

$$\phi(r, \theta) = \sum_{n=0}^{\infty} b_n(\theta) F_n(r)$$

$$= b_0(\theta) F_0(r) + \sum_{n=1}^{\infty} b_n(\theta) F_n(r)$$

$$\downarrow \quad \downarrow$$

$$= (a_0 \cdot b_0)$$

$$\downarrow$$

$$\underline{A_0/2} \equiv a_0 \cdot b_0 + \sum \dots$$

$$= \frac{A_0}{2} + \sum_{n=1}^{\infty} (\hat{A}_n \cos(n\theta) + \hat{B}_n \sin(n\theta)) C_n r^n$$

$$A_n = \hat{A}_n C_n$$

$$B_n = \hat{B}_n C_n$$

$$\phi(r, \theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} r^n (A_n \cos(n\theta) + B_n \sin(n\theta))$$

Final B.C.  $\phi(1, \theta) = f(\theta)$

$$\phi(1, \theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} (A_n \cos(n\theta) + B_n \sin(n\theta)) = f(\theta)$$

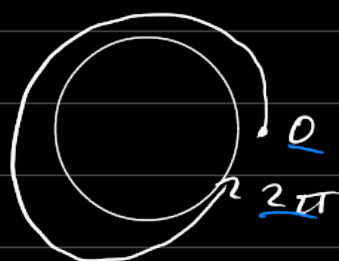
• Can solve as before:

$$\int_0^{2\pi} \underbrace{\cos(n\theta) \sin(n\theta)}_{\text{odd } f^n} d\theta \rightarrow 0$$

\*  $A_n, B_n, A_0$ : Fourier coeff. of  $f(\theta)$

Recall:  $\int_0^L$  (before)

(now)



$$\int_0^{2\pi}$$

